

CHURCHILL FALLS, NL, Canada

WMO: 710020

Lat: 53.562N

Lon: 64.106W

Elev: 440

StdP: 96.16

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.6	(c) -32.4	(d) -39.6	(e) 0.1	(f) -34.4	(g) -37.5	(h) 0.1	(i) -32.2	(j) 11.6	(k) -17.6	(l) 10.4	(m) -16.8	(n) 2.1	(o) 270	(p) 0.776

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.6	(c) 25.0	(d) 16.1	(e) 22.8	(f) 15.1	(g) 20.8	(h) 14.3	(i) 17.7	(j) 22.4	(k) 16.5	(l) 20.7	(m) 15.5	(n) 19.2	(o) 4.2	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	12.1	18.9	14.9	11.2	18.0	13.8	10.4	17.1	51.9	22.3	48.1	20.7	44.9	19.2	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.2	(b) 9.0	(c) 8.0	(e) -38.5	(f) 28.7	(g) 2.2	(h) 1.5	(i) -40.1	(j) 29.8	(k) -41.3	(l) 30.7	(m) -42.6	(n) 31.6	(o) -44.1	(p) 32.7		
(4) 10.2	(5) 9.0	(6) 8.0	(7) -38.6	(8) 19.8	(9) 2.2	(10) 1.4	(11) -40.1	(12) 20.7	(13) -41.4	(14) 21.5	(15) -42.6	(16) 22.3	(17) -44.2	(18) 23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.1	-20.4	-19.5	-12.7	-4.2	4.1	10.6	14.4	13.5	8.1	1.7	-6.6	-15.3
	DBStd	13.40	7.58	7.18	7.36	5.78	4.53	4.37	3.26	3.29	3.82	3.98	5.63	7.32
	HDD10.0	4763	943	827	704	425	192	43	4	6	80	258	497	784
	HDD18.3	7467	1201	1060	963	675	442	236	127	153	306	515	747	1043
	CDD10.0	350	0	0	0	0	8	60	141	116	24	1	0	0
	CDD18.3	14	0	0	0	0	0	3	7	4	1	0	0	0
	CDH23.3	146	0	0	0	0	6	37	57	41	6	0	0	0
	CDH26.7	19	0	0	0	0	1	8	7	3	1	0	0	0
Wind	WSAvg	3.7	3.8	3.8	4.2	4.2	3.7	3.7	3.3	3.2	3.7	3.9	3.7	3.6
Precipitation	PrecAvg	999	72	58	69	70	60	98	117	96	115	85	89	71
	PrecMax	1234	130	122	161	119	107	184	205	195	154	152	155	158
	PrecMin	808	26	14	30	31	27	36	44	28	48	47	24	36
	PrecStd	120	33	25	33	27	25	46	38	41	29	25	33	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.3	1.9	5.7	11.8	23.3	27.4	27.4	26.9	23.4	15.5	8.3	2.7
		MCWB	0.6	0.9	1.1	4.7	13.0	16.3	17.2	17.9	16.2	11.3	7.4	1.8
	2%	DB	-1.1	-0.5	2.7	8.3	18.9	23.6	24.8	24.3	19.2	12.1	4.8	0.7
		MCWB	-1.8	-1.4	0.4	3.0	10.1	14.4	16.2	16.8	13.9	9.8	3.8	0.2
	5%	DB	-4.5	-4.2	0.7	6.0	15.2	20.8	22.8	21.7	16.6	9.8	2.4	-1.2
		MCWB	-5.1	-5.2	-1.0	2.2	8.2	13.3	15.1	15.5	12.4	7.6	1.4	-1.8
	10%	DB	-8.3	-7.8	-1.3	4.1	12.1	18.5	20.8	19.7	14.5	7.7	0.7	-4.4
		MCWB	-8.9	-8.6	-3.0	1.2	6.9	12.2	14.6	14.6	11.1	6.0	0.0	-5.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	1.2	2.6	5.8	13.5	18.1	19.4	19.6	17.3	12.1	7.5	2.2
		MCDB	1.0	1.8	4.4	9.1	21.4	24.2	22.2	24.0	22.3	13.7	8.1	2.7
	2%	WB	-1.7	-1.4	0.6	4.0	10.9	16.0	17.5	17.9	14.9	10.1	3.9	0.3
		MCDB	-1.0	-0.6	2.2	7.0	16.8	21.0	21.9	22.3	17.9	11.9	4.5	0.7
	5%	WB	-5.0	-5.2	-1.0	2.7	9.1	14.5	16.5	16.4	13.3	8.0	1.5	-1.9
		MCDB	-4.4	-4.1	0.9	5.6	13.9	18.9	20.7	19.9	16.3	9.2	2.2	-1.3
	10%	WB	-8.8	-8.6	-2.9	1.5	7.2	13.1	15.6	15.1	11.5	6.0	0.2	-5.1
		MCDB	-8.3	-7.8	-1.3	4.1	12.0	17.2	19.4	18.5	13.8	7.4	0.7	-4.4
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.3	12.1	10.3	9.9	10.3	9.6	8.8	8.1	5.9	6.8	8.6
		MCDBR	10.4	10.8	11.5	12.7	16.6	14.5	13.2	12.7	12.3	10.1	7.7	8.3
		MCWBR	10.1	10.2	9.6	9.0	9.7	7.5	6.5	6.8	8.1	7.8	7.2	8.4
	5% WB	MCDBR	10.3	10.7	10.5	10.4	14.5	11.5	10.4	10.4	10.9	8.8	7.4	8.4
		MCWBR	10.1	10.1	9.1	7.8	9.1	6.8	6.2	6.2	7.9	7.3	7.1	8.4
Clear-Sky Solar Irradiance	taub	0.233	0.246	0.268	0.315	0.355	0.360	0.387	0.353	0.331	0.295	0.253	0.232	
	taud	2.233	2.246	2.221	2.163	2.297	2.419	2.366	2.471	2.499	2.532	2.305	2.174	
	Ebn at Noon	761	872	923	913	892	890	856	870	848	814	727	672	
	Edh at Noon	59	82	106	130	121	109	113	96	82	64	56	52	
All-Sky Solar Radiation	RadAvg	1.01	2.11	3.57	4.92	5.31	5.12	4.71	3.81	2.82	1.57	0.80	0.67	
	RadStd	0.08	0.18	0.32	0.38	0.46	0.39	0.29	0.24	0.26	0.11	0.10	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	+1.16	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page